



INFORMATION SHEET

GROUP DESIGN PROJECT
CRANFIELD UNIVERSITY

AEGIS

ASTEROID EXPLORATION & GRAVITATIONAL INTERIOR SURVEYOR



WORK BREAKDOWN STRUCTURE

SALFORD & CO.

WP1000: SYSTEMS

- TARA
- FINN

WP2000: MISSION

- PAU M
- YOGESH

WP3000: MECHANICAL

- PAU C

WP4000: ELECTRICAL

- MIKOLAJ
- VIGNESH

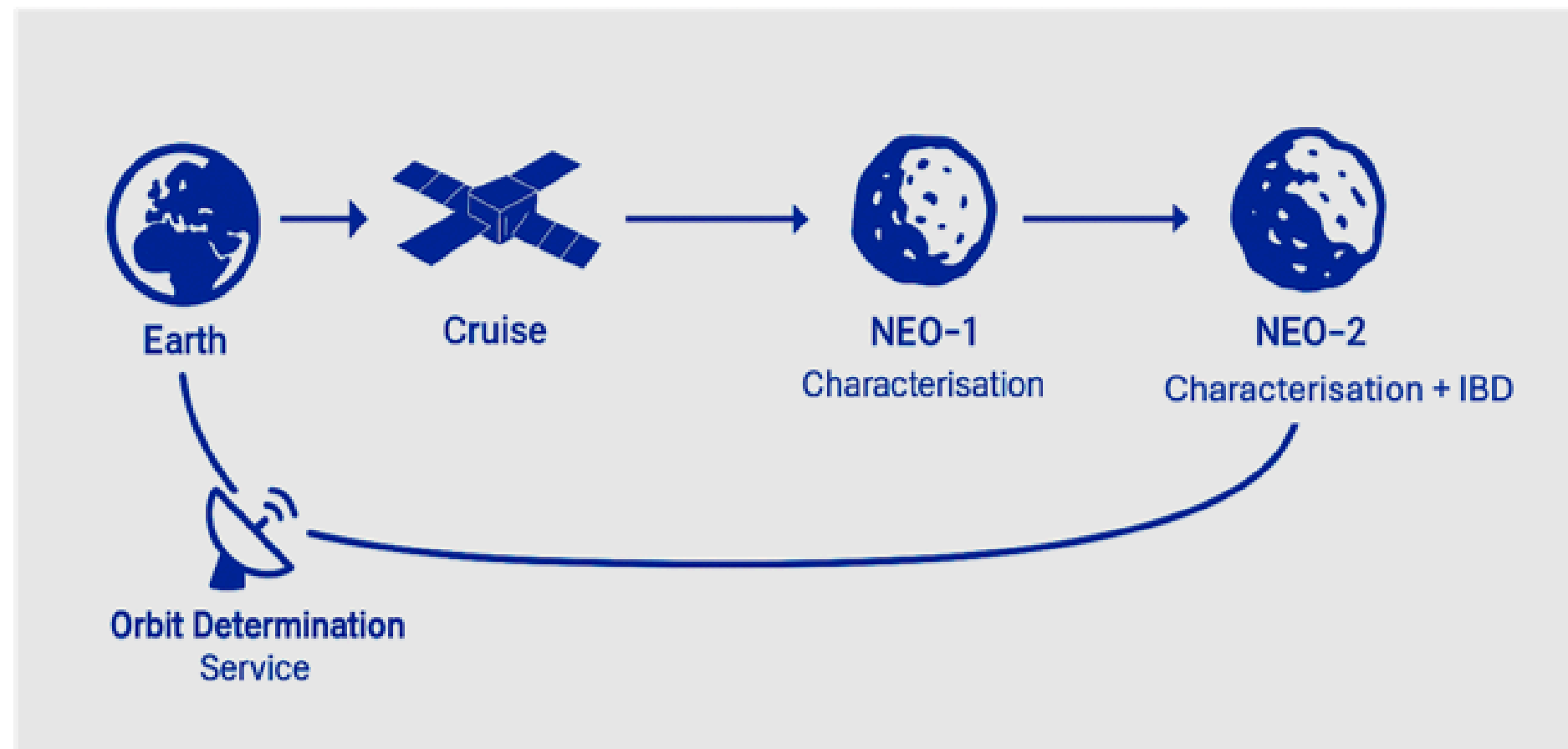
WP5000: PAYLOAD

- JAMES
- JACK
- GERARD
- MAEL
- SERGIO

Mission Overview

“Investigate the feasibility of a mission that will rendezvous with multiple near-Earth asteroids (NEOs) to refine their orbital & physical properties, while demonstrating a controlled mitigation technique.

In parallel, develop an independent orbit-determination service that utilises the spacecraft-acquired data.”



Mission Overview

****INSERT**

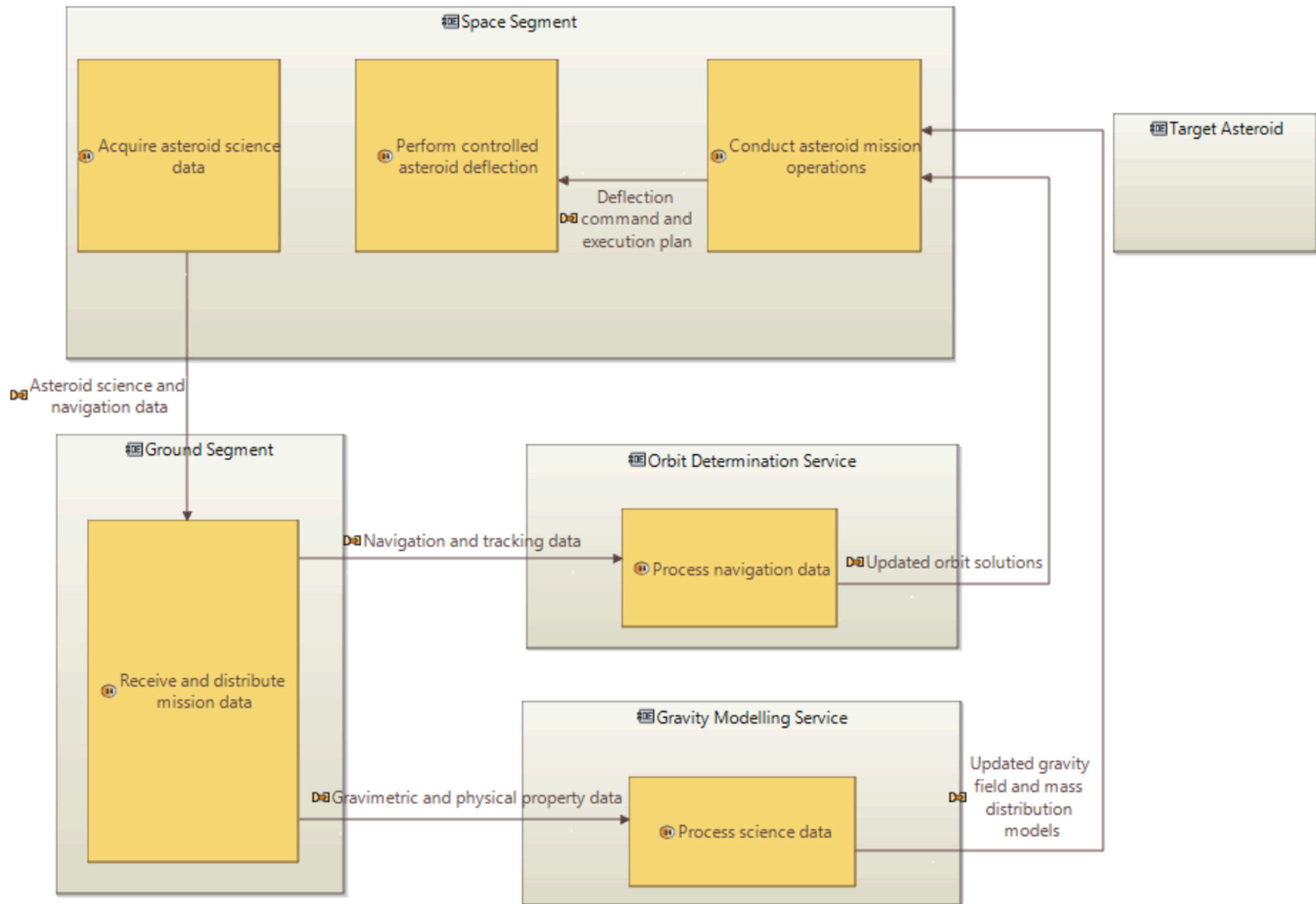
CAD

MODEL*

****INSERT**

TRAJECTORY

GRAPHIC *



CONOPS

****INSERT CONOPS
GRAPHIC ***

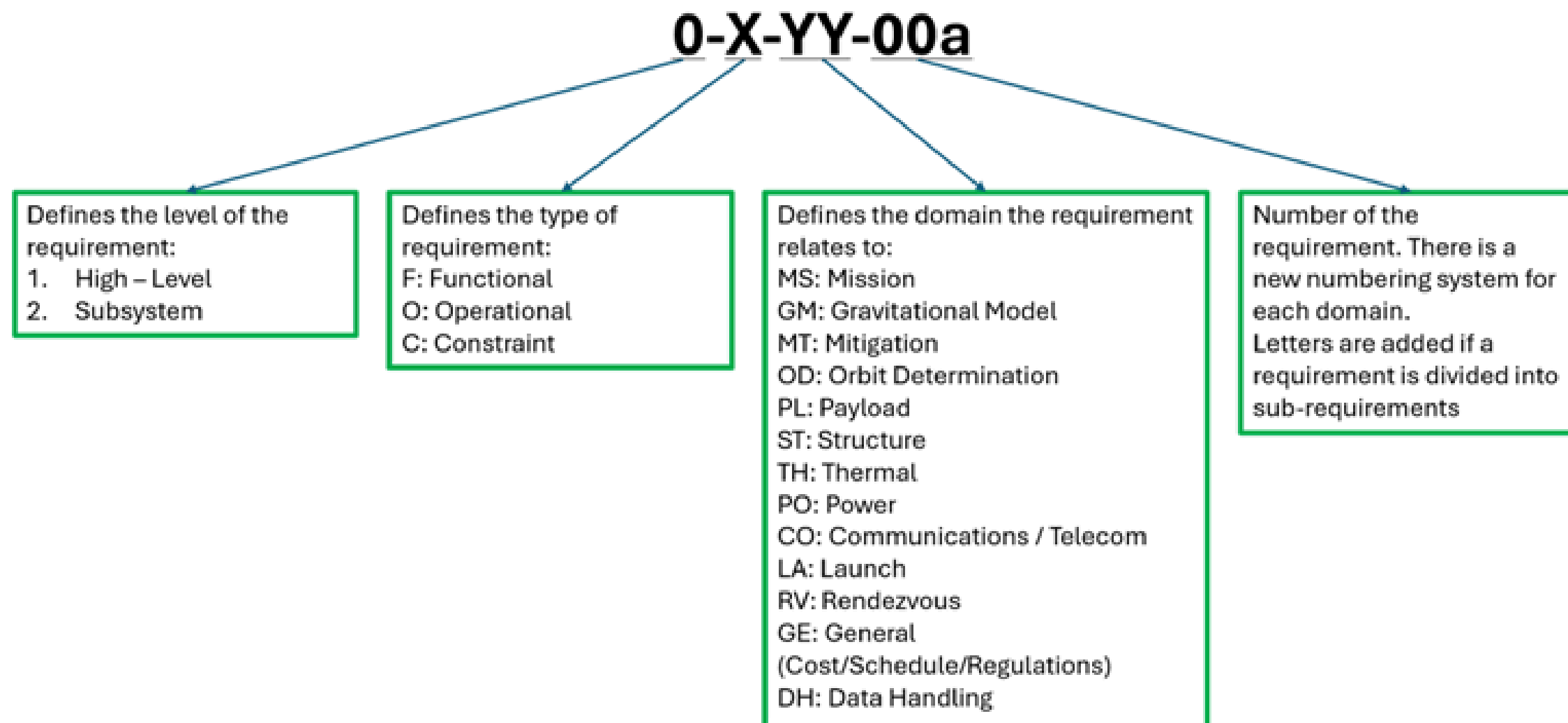
HIGH LEVEL REQUIREMENTS

ID	Type	Requirement
REQ-F-001	Functional	The mass and bulk density of the target NEO shall be determined to an accuracy of TBD% using gravimetric measurements collected over TBD days
REQ-F-002	Functional	The spacecraft shall image the surface of each target asteroid with a camera capable of TBD spatial resolution at TBD km
REQ-F-003	Functional	The scanning LIDAR should be able to produce at least 1 range point per square meter across the surface.
REQ-F-004	Functional	The mission shall quantify discrepancies between measured and modelled gravity fields to validate our predictions
REQ-F-005	Functional	The spacecraft shall demonstrate a controlled mitigation technique on the final target asteroid, deflecting it by 500km via a total ΔV change of 1mm/s.
REQ-F-006	Functional	An independent ground-based orbit-determination system shall be developed with the capability to estimate trajectories of NEOs with a position uncertainty of $\pm$ TBD km

HIGH LEVEL REQUIREMENTS

ID	Type	Requirement
REQ-O-007	Operational	The mission operational duration shall be no less than TBD years
REQ-O-008	Operational	Spacecraft shall support a data downlink rate no less than TBD Mbps over TBD contact windows
REQ-O-009	Operational	The collected data shall be timestamped and contain data identifying the location and movements of the detected NEO(s)
REQ-O-010	Operational	Spacecraft shall transmit data collected within TBD s/hrs of detection
REQ-O-011	Operational	The mission shall conduct sequential, data-collecting campaigns at both NEOs
REQ-O-012	Operational	The orbit determination (OD) model shall detect and ingest new MPC data every hour.
ID	Type	Requirement
REQ-C-013	Constraint	The total mission budget shall not exceed 550 million EUR
REQ-C-014	Constraint	The spacecraft shall be designed, manufactured and tested within a period not exceeding 4 years
REQ-C-015	Constraint	The mission shall target two near earth asteroids of varying size or composition
REQ-C-016	Constraint	The OD model shall require TBD GB of RAM, TBD GB of storage and TBD CPU cores.

SUBSYSTEM REQUIREMENTS



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****INSERT**

**SUBSYSTEM
REQUIREMENTS***